

Kabaro Valley Smallholder Irrigation Pilot Final Report

Kabaro Valley Agricultural Research Consortium

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NOTE: This is a fictional document created as sample data. The Kabaro Valley, the consortium and all figures within are invented and must not be cited as real research.

Abstract

This report presents the findings of a 14-month controlled pilot comparing three irrigation approaches across 72 smallholder plots in the Kabaro Valley. The pilot tested whether low-cost drip irrigation, paired with a shared scheduling service, could raise yields without increasing total water withdrawal.

Plots using drip irrigation with scheduling recorded a mean yield increase of 34 percent against the control, while reducing water applied per hectare by 22 percent. Drip irrigation without scheduling produced a smaller yield gain of 19 percent and no meaningful reduction in water use. The difference between the two treatment arms was statistically significant at the 5 percent level.

The pilot concludes that the scheduling component, rather than the irrigation hardware alone, accounts for most of the water saving. Recommendations for a wider rollout appear in section 7.

1. Background

The Kabaro Valley supports approximately 4,100 smallholder households farming plots with a median size of 0.6 hectares. Rainfall is bimodal but has become less predictable over the past decade, and the shorter of the two rainy seasons now fails in roughly one year in three.

Most households irrigate by furrow from shallow wells or from the Kabaro river during the dry months. Furrow irrigation is labour-intensive and applies water unevenly, with losses concentrated at the head of the field.

2. Objectives

The pilot was designed to answer three questions:

- Does low-cost drip irrigation raise yields on plots of this size and crop mix?
- Does a shared scheduling service, delivered by SMS, change how much water is applied?
- Are the capital and running costs recoverable within three growing seasons at prevailing farm-gate prices?

A fourth question, on whether shared equipment ownership between households is viable, was dropped after the design phase because the sample would have been too small to detect an effect.

3. Site selection

Twelve sub-districts were assessed against soil type, water access and distance to the nearest market. Four were excluded for lack of reliable dry-season water and one for a competing programme already operating there. The remaining seven contributed households to the sample.

4. Methodology

4.1 Design

The pilot used a three-arm randomised design running from September 2023 to October 2024, covering two full growing seasons. Households were randomised at the plot level within sub-district blocks to control for local water availability.

Arm	Treatment	Plots
A (control)	Furrow irrigation, no scheduling	24
B	Drip irrigation, no scheduling	24
C	Drip irrigation plus SMS scheduling	24

4.2 Measurement

Water applied was measured with inline flow meters read fortnightly by enumerators. Yield was measured at harvest by weighing the marketable output of a 50 square metre sub-plot at the centre of each field, then scaling to the full plot area.

Labour hours were self-reported in a weekly diary. Because self-reporting is unreliable, labour findings are treated as indicative only and are not used in the cost-recovery calculation in section 6.

4.3 Attrition

Five households left the pilot before completion: three relocated, one withdrew after a dispute over well access, and one was excluded after the flow meter was found to have been bypassed. Attrition was balanced across arms and the analysis uses the 67 plots that completed both seasons.

5. Results

5.1 Yield

Yields are reported in kilograms per hectare, averaged across both seasons and across the crop mix, which was predominantly onion, tomato and leafy greens.

Arm	Season 1	Season 2	Mean	Change vs control
A (control)	8,410	8,190	8,300	—
B	9,720	10,050	9,885	+19.1%
C	10,940	11,320	11,130	+34.1%

The gap between arms B and C widened in the second season, which the field team attributes to households becoming more confident in following scheduling messages that initially contradicted their own judgement.

5.2 Water applied

Arm	Cubic metres per hectare	Change vs control
A (control)	4,620	—
B	4,480	-3.0%
C	3,600	-22.1%

Arm B applied almost as much water as the control despite the more efficient hardware. Enumerator notes suggest households in arm B ran the system on the same schedule they had used for furrow irrigation, so the efficiency gain was absorbed by over-watering rather than converted into a saving.

6. Costs and recovery

Capital cost per plot covers the tank, filter, mainline and driplines, sized for 0.6 hectares. The scheduling service cost includes SMS delivery and the share of the agronomist's time attributable to the pilot.

Item	Arm B	Arm C
Capital cost per plot	31,500	31,500
Annual running cost	2,900	4,700
Additional revenue, year 1	14,200	25,400
Seasons to recover capital	3.4	1.9

All monetary figures are in local currency units at October 2024 prices. Arm C recovers its capital cost within two growing seasons at prevailing farm-gate prices, comfortably inside the three-season threshold set at the design stage. Arm B does not.

7. Limitations

The pilot ran during two seasons of near-average rainfall. Performance under a failed short rains season, which is the scenario of greatest concern, was not observed and cannot be inferred from these results.

The sample is drawn from households that volunteered, who may be more receptive to new practice than the district as a whole. The yield measurement method assumes the central sub-plot is representative, which tends to overstate yields on fields with uneven water distribution — an effect that would inflate the control arm more than the treatment arms, making the reported gains conservative.

Farm-gate prices were unusually stable across the pilot period. A price fall of more than 15 percent would push arm C beyond the three-season recovery threshold.

8. Recommendations

The consortium makes four recommendations.

- Any wider rollout should bundle scheduling with hardware. Distributing drip equipment alone is not supported by these findings, since arm B delivered under half the yield gain and no water saving.
- The scheduling service should be tested against a failed short rains season before commitments are made beyond the pilot districts.
- Flow meters should be retained in the first year of any rollout. The single bypass incident suggests metering also serves a monitoring function that self-reporting cannot replace.
- A follow-on study should examine shared ownership between two or three households, which was dropped from this pilot for sample size reasons but remains the most plausible route to affordability for the smallest plots.

9. Acknowledgements

The consortium thanks the participating households, the enumerator team, and the district agricultural office. Responsibility for the analysis and for any errors rests with the consortium alone.

10. Data availability

Anonymised plot-level data is available on request to the consortium secretariat. Requests are reviewed within 30 days. Raw enumerator diaries are not released because they contain identifying detail that cannot be reliably removed.